Name: brightlearn-de-exam-mpho-matebesi
Course: Data Engineering
Project Name: BrightLearn Data Warehouse Build
Project Code: BL-DE-EXAM-2026-07

BrightMart: Data Quality Findings Report

1. Overview

This report breaks down the specific data quality issues flagged within BrightLearn_Raw_Data.csv. Our goal here is simple: clean up and standardise this dataset before we start building out the ETL pipeline and data warehouse. Left unchecked, these anomalies will skew our analytics and break pipeline logic. Below is a detailed breakdown of what we found, how many records are impacted, and how we plan to fix them.

2. How We Analysed the Data

We ran the source CSV through a mix of target SQL queries in SSMS alongside quick spreadsheet profiling. We specifically targeted:

- Blank or missing fields
- Duplicate entries
- Clashing date and text formats
- Out-of-bounds numbers and logical errors
- Outliers and cross-column mismatches

3. Detailed Breakdown of Findings

Messy Date Formats

- The Issue: Transaction dates are all over the place. We found a mix of YYYY-MM-DD, DD/MM/YYYY, DD-MM-YYYY, and even text strings like 23 Apr 2024.
- Impact: 45 rows affected.
- The Fix: Force everything into standard ISO format (YYYY-MM-DD) during the staging phase.

Negative Amounts

- The Issue: We spotted several transactions with negative values (e.g., -40.02, -201.0).
- Impact: 30 rows affected.
- The Fix: These are either typos or actual refunds. We will split them: isolate and flag true refunds, and drop or fix pure input errors.

Impossible Discounts

- The Issue: Some discount fields are higher than the actual transaction amount (like a R500 discount on a R299 item).
- Impact: 20 rows affected.
- The Fix: Cap the discount at the total transaction value for now and flag these specific rows for the business team to look at.

Blank Customer Profiles

- The Issue: Critical gaps in customer_email, customer_phone, and customer_city. In several spots, the entire customer profile is blank.
- Impact: 50 rows affected.
- The Fix: If a row has zero customer identifiers, we drop it. For minor missing fields, we will fill blanks with an "Unknown" string.

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Ghost Duplicates

- The Issue: Exact duplicate rows where the customer, date, and product match perfectly.
- Impact: 40 rows affected.
- The Fix: Deduplicate the set so we don't accidentally double-count revenue.

Inconsistent Text Casing (Payment Methods)

- The Issue: Payment types use messy, mixed casing and naming (e.g., Cash, CASH, DEBIT CARD, Debit Card).
- Impact: 25 rows affected.
- The Fix: Standardise everything into clean categories: Cash, Debit Card, Credit Card, EFT, or Store Credit.

Missing Product Categories

- The Issue: Items like "Men's Running Shorts" are listed, but their main category or subcategory fields are completely blank.
- Impact: 15 rows affected.
- The Fix: Map these back to their correct category using the SKU, and fill in the missing text.

Massive Transaction Spikes

- The Issue: A few transaction totals are massive (e.g., 10496.0, 7047.0) compared to ordinary day-to-day retail shopping.
- Impact: 10 rows affected.
- The Fix: Keep them in the system but flag them as bulk corporate buys. Pull them out of our everyday average calculations so they don't skew the data.

Broken Inventory Logic

- The Issue: Items where stock on hand has fallen below the reorder threshold, yet no automatic reorder was triggered.
- Impact: 20 rows affected.
- The Fix: Flag these immediately for the inventory management team to reconcile.

4. Summary & Next Steps

The raw data requires significant cleanup before being pushed straight into production. Before writing the ETL code, we need to run a pre-processing script to standardise formats, strip out duplicates, fill in missing variables, and isolate the outliers for business review. Taking these steps ensures our final data warehouse rests on a reliable, clean foundation.

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Harvard Bibliography References

- Batini, C. & Scannapieco, M. (2016). *Data Quality: Concepts, Methodologies and Techniques.* Springer.
- Kimball, R. & Ross, M. (2013). *The Data Warehouse Toolkit: The Definitive Guide to Dimensional Modeling.* Wiley.
- Redman, T.C. (2013). *Data Driven: Profiting from Your Most Important Business Asset.* Harvard Business Review Press.
- Wang, R.Y. & Strong, D.M. (1996). 'Beyond Accuracy: What Data Quality Means to Data Consumers.' *Journal of Management Information Systems*, 12(4), pp. 5–33.